

Bed-Vacancy Alarm System

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This inexpensive, compact bed-vacancy alarm system raises an audible alarm whenever a patient or a small kid sleeping on the bed leaves the bed.

The circuit comprises a bed sensor, an alarm box and a power supply with an optional battery-backup facility. The bed sensor can be placed under the hip of the care receiver on the mattress. The alarm box can be hooked onto a headboard/nightstand.

The highly-sensitive sensor used is nothing but a force sensing resistor (FSR). The FSR is a polymer thick-film (PTF) device whose resistance decreases with an increase in the force applied to its active surface. It is not a strain gauge, load cell or pressure transducer.

While FSR can be used for dynamic measurement, only qualitative results are generally obtainable. Usually an FSR can sense applied force anywhere in the range of 100gm to 10kg; when no pressure is applied, its resistance is larger than 100kΩ.

PARTS LIST

Semiconductors:

- IC1 - ATmega328 MCU with Arduino UNO bootloader
- IC2 - LM117-5 LDO linear regulator
- D1, D2 - 1N5819 Schottky barrier rectifier
- LED1, LED2 - 5mm LED
- T1 - BC547 npn general-purpose transistor

Resistors (all 1/4-watt, ±5% carbon):

- R1, R2 - 10-kilo-ohm
- R3, R5 - 1-kilo-ohm
- VR1 - 100-kilo-ohm preset

Capacitors:

- C1 - 47µF, 25V electrolytic
- C2 - 47µF, 16V electrolytic
- C3, C6 - 100nF ceramic disk
- C4, C5 - 22pF ceramic disk

Miscellaneous:

- SW₁ - 16MHz crystal
- S1 - SPDT switch
- S2 - Tactile switch
- BAT - 9V Battery
- FSR1 - FSR interface
- PZ1 - Piezo buzzer interface
- J1 - Connector for (9V-12V) DC supply
- J2 - Connector for FTDI basic board interface

At the heart of the alarm box is 28-pin DIP AVR microcontroller (MCU) ATmega328P-PU (IC1). It can be programmed with embedded software using a standard programmer. ATmega328P-PU offers 23 input/output functional ports; a 16MHz crystal oscillator is used to provide timing/clock reference.

The power supply converts the DC from an AC mains adaptor (9V-12V/1A) into a stable DC of 5V. The regulated voltage supplied by linear voltage regulator LM117-5.0 (IC2) is used to power the circuit.

IC2 is a low-dropout voltage regulator with a dropout of 1.2V at 800mA of load current. Available in packages like TO-220 and SOT-223, it offers current-limiting and thermal shutdown. The optional battery-backup can be from a 9V (PP3/6F22) battery pack.

FSR integration

Usually size and shape are the limiting factors in an FSR (Fig. 1) integration. Response of FSR is sensitive to the distribution of the applied force. It is therefore necessary to provide equal force distribution, using a firm, flat and smooth mounting surface.



Fig. 1: A force sensing resistor

An FSR typically has an air vent that runs from the open active area down the length of the tail and out to the atmosphere. This vent assures pressure equilibrium with the environment, and allows

loading and unloading of the device. Blocking this vent could cause the FSR to respond erratically. Entire surface of the sensor should be covered using thin, uniform double-sided laminating adhesive.

Never try to solder directly to naked silver traces. A suitable zero insertion force (ZIF) style connector is recommended for interconnection.

Circuit and working

After wiring the circuit as per Fig. 2, place the bed sensor (FSR) under the mattress and switch on using S1.

Set VR1 to its mid-travel (50k) and measure DC voltage at TP3 using a digital multimeter. Note down DC voltage readings when the bed is vacant or occupied. Based on this, you can add your own bed-occupancy and bed-vacancy threshold values in the software, with the help of the following relationship:

$$TH = [1023/5] \times V$$

where TH is the threshold value and V is the DC voltage at TP3.

For example, if the DC voltage at TP3 is 4, when the bed is occupied, bed-occupancy threshold will be $[1023/5] \times 4 = 818$.

Next, keep the system in power-off mode and upload the finished code to the MCU using an Arduino UNO board. The driving transistor (T1) selection depends on the piezo speaker used.

Construction and testing

An actual-size, single-side PCB for the bed-vacancy alarm system is shown in Fig. 3 and its component layout in Fig. 4. Assemble the circuit on PCB to save time and minimise assembly errors.

Use a proper IC base for the MCU. The prototype was tested using a 1.27cm (0.5-inch) dia FSR bought from www.rhydolabz.com. The FSR can safely handle a maximum applied force of about 10kg only. So if you want to make bed-vacancy alarm system for



TABLE I
Test Points

Test point	Details
TP0	GND
TP1	5V
TP2	5V in working condition
TP3	FSR output voltage